Method Validation Report

The Validation of an ECL Assay for the Detection of Anti-Drug Antibodies (ADA) Directed Against LT3114 in Monkey Serum

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Client Project Number: NA

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METHOD VALIDATION REPORT

REPORT TITLE:

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QPS PROJECT NUMBER:

344-1404

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1. SIGNATURE PAGE

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2. COMPLIANCE STATEMENT

This portion of the study was conducted in compliance with the Food and Drug Administration (FDA) Good Laboratory Practice Regulations (GLP) as set forth in Title 21 of the U.S. Code of Federal Regulations Part 58 and the FDA Guidance for Industry: Bioanalytical Method Validation, May 2001.

To the best of my knowledge this report accurately reflects the raw data. I have reviewed the study and agree that the data supports the conclusions stated herein.

Breann Barker
Breann Barker, Ph.D.
Principal Investigator
QPS, LLC

8/20/14
Date

3. QUALITY ASSURANCE STATEMENT

QPS: 344-1404

Protocol Title: The Validation of an ECL Assay for the Detection of Anti-Drug Antibodies (ADA) Directed Against LT3114 in Monkey Serum

This QPS study has been audited and found to accurately reflect the requirements set forth by the validation plan 344-1404 and standard operating procedures and the results reported accurately reflect the source data. The following table summarizes the inspections and audits performed throughout the course of the study and the dates when each audit report and any findings were reported by QPS Quality Assurance.

Audit Description	QPS Auditor(s)	Audit Dates	Date Audit Findings Reported to QPS Principal Investigator and Management
Draft Validation Plan	S. Ma	2 Jul 2014	2 Jul 2014
Study Records/ Draft Report	R. McCrae	28-31 Jul 2014 1, 4 Aug 2014	4 Aug 2014

The undersigned has assured that the QPS final report accurately reflects the source data.

Rita McCrae

Rita McCrae, M.S.
QPS Quality Assurance

8/19/14
Date

4. VALIDATION SUMMARY FOR ANTI-LT3114 ANTIBODIES

A quasi-quantitative immunogenicity assay to detect the presence of anti-LT3114 antibodies in monkey serum samples was validated. Experiments were conducted at QPS to assess assay precision, screening and confirmation assay cut-point factor for monkey serum, matrix selectivity, drug tolerance, and PC stability. The results are provided in the following summary table.

Report Title	The Validation of an ECL Assay for the Detection of Anti-Drug Antibodies (ADA) Directed Against LT3114 in Monkey Serum		
Report Number	QPS 344-1404		
Analyte Name	Anti-LT3114 antibodies in monkey serum		
Negative Control (NC)	A pool of monkey serum prepared from at least 30 individual lots		
Positive Controls (PC)	Whole LT3114 anti-idiotype rabbit positive serum		
Minimum Required Dilution (MRD) and Sample Volume:	MRD = 1:30 (prior to addition to master mix) Minimal 10 μL for screening Minimal 105 μL for confirmation		
Analytical Method Type	ECL		
Detection Instrument	MSD Sector Imager 6000		
Analytical System Software and Version	Discovery Workbench^{TM} 2006 MSD 3.0.17_3		
Inter-Run Precision (%CV) of Anti-LT3114 positive rabbit serum (HPC 1:500, MPC 1:2000 & LPC 1:6000 in monkey serum pool)	HPC	MPC	LPC
	11.4%	9.8%	9.0%
Intra-Run Precision (%CV) of Anti-LT3114 positive rabbit serum (HPC 1:500, MPC 1:2000 & LPC 1:6000 in monkey serum pool)	HPC	MPC	LPC
	4.3%	4.4%	3.4%
Cut Point Factor (Normalization factor) for screening	1.18		
Calculation for Plate Cut Point	Mean NC * 1.18		
Confirmatory Cut Point Factor	38.0%		

Room Temperature Stability of PC in Monkey Serum	24 hours at room temperature
Freeze/Thaw Stability of PC in Monkey Serum	5 Cycles at -70°C
Drug Tolerance	HPC (1:500) can tolerate at least 100 μg/mL free LT3114 and LPC (1:6000) can tolerate at least 4 μg/mL free LT3114 in monkey serum
Matrix Selectivity (Recovery)	HPC (1:500): 100% lots tested have recovery between 75% and 125% LPC (1:6000): 90% lots tested have recovery between 75% and 125%

5. OBJECTIVE

The purpose of this study was to validate an ECL method to detect the presence of anti-LT3114 in monkey serum samples.

6. ABBREVIATIONS AND DEFINITIONS

This section provides abbreviations and definitions of terms and concepts that may be commonly used throughout this report.

ADA	Anti-Drug Antibody
PC	Positive Control
HPC	High Positive Control
LPC	Low Positive Control
MRD	Minimum Required Dilution
CPF	Cut Point Factor
MKSE	Monkey Serum
MSD	Meso Scale Discovery
pAb	Polyclonal Antibody
NA	Not Applicable
NC	Negative Control
NR	Not Reported
PC	Positive Control
RU	Response Unit
SD	Standard Deviation
SOP	Standard Operating Procedure
Precision	$\%\text{Coefficient of Variation (\%CV)} = 100 \times \dfrac{\text{Standard Deviation}}{\text{Mean Value}}$

One pooled monkey serum sample was used as negative control (NC) for each validation run. Whole LT3114 anti-idiotype rabbit positive serum was diluted in monkey serum at 1:500, 1:2000, and 1:6000 and used as HPC, MPC and LPC, respectively, in this qualitative assay.

All statistics in the data tables are calculated by Watson IRM 7.4.1 or based on the "precision as displayed" option of Microsoft® Excel (e.g., Mean, SD, %CV, %Diff).

7. ACCEPTANCE CRITERIA

The validation acceptance criteria and statistical data will be determined at a minimum per QPS 344-1404 validaton protocol and QPS-TLM-001, if applicable. Each control sample will be diluted in the same manner analogous to how unknowns will be treated. Samples and PCs are added to duplicate wells.

- For NCs, PCs and samples, the %CV from duplicate measurements should not be greater than 25%. Samples with a duplicate %CV greater than 25% will be rejected and will not be included in statistical analyses.
- At least two-thirds of NCs should be reportable.

- The mean values of the ECL responses of NC and three levels of PCs must be in the order of: NC< LPC< MPC <HPC.
- For the plate to be accepted at least 50% of the PCs at each level should be reportable and at least two-thirds of all PCs should be reportable.

Method acceptance criteria:

Intra-assay precision: The intra-assay precision will be determined from the NC and the PCs on the assay plate. The mean and %CV of the response unit (RU) from the PCs and NC will be evaluated. The %CV of the NC and PCs at each level must be ≤ 25%.

Inter-assay precision: The inter-assay precision will be determined from all acceptable validation runs except intra-assay precision runs. The mean and %CV of the response unit (RU) from the PCs and NC will be evaluated. The %CV of the NC and PCs at each level must be ≤ 25%.

Matrix Specificity (Recovery): The % recovery of the mean response of PC-spiked individual lots will be assessed. At least, two-thirds of the tested lots should be within the acceptable range (75-125% recovery).

Drug tolerance test: Observations will be made as to which concentration of drug inhibits the ability of the assay to detect the PC at each level.

Freeze/thaw stability: The samples are considered stable if the mean RU value of 5 freeze/thaw cycle samples are within ± 25% difference of the mean RU value of PCs that were thawed once immediately before analysis.

Room temperature stability: The samples are considered stable if the mean RU value of the samples maintained at room temperature for 24 hours is within ± 25% difference of the mean RU value of untreated PCs.

8. MATERIALS

8.1. Sample-Handling Equipment

Equivalent equipment may be substituted on an as-needed basis
Plate Sealer: Costar Cat.# 3095
25-100 mL Reagent Reservoir: Costar Cat. # 4870
96-well format micro titer/dilution tubes (VWR Cat. # 46300-536)
Polypropylene dilution plate (Nunc Cat. # 267385)
0.5-50 mL polypropylene tubes, VWR Scientific
500 or 1000 mL wash bottle
Graduated Cylinder
Multi-tube vortexer: VWR Scientific Products
Plate Shaker
Microplate reader: Meso Scale Discovery Sector Imager 6000
Balance

8.2. Analytical Equipment

Microplate reader: Meso Scale Discovery Sector Imager 6000

9. METHODS

9.1. Preparation of Negative and Positive Controls

Pooled normal monkey serum was used as the negative control. Whole LT3114 anti-idiotype rabbit positive serum (Lot# R-2265-B2-11/13) was diluted 1:500, 1:2000, and 1:6000 in the pooled negative control. Positive control working stock solutions was prepared in bulk, aliquoted and stored at -70°C.

9.1. Critical Reagents

Name:	LT3114 (HU-anti-LPA antibody)
Source:	LPath, Inc.
Concentration:	19.33 mg/mL (Purity applied)
Lot Number:	LP121416
Expiration Date:	21-May-2016
Storage Conditions:	-70°C

See CoA in Appendix A

9.2. Positive Control Source Information

Name:	Whole LT3114 anti-idiotype rabbit positive serum
Source:	LPath, Inc.
Concentration:	Unknown
Lot Number:	R-2265-B2-11/13
Expiration Date:	21-May-2016
Storage Conditions:	-70°C

See CoA in Appendix B

9.3. Analytical Method

Parameter	Value of Parameter
General:	
Analyte	LT3114 Antibody
Matrix	Monkey Serum
Control Concentrations	NC: pooled monkey serum PC: Whole LT3114 anti-idiotype rabbit positive serum was diluted in monkey serum at 1:500, 1:2000, and 1:6000
Minimum Required Dilution (MRD)	30

Sample Volume	10 µL for screening 105 µL for confirmation
Assay Procedure Summary	In the assay, monkey serum samples, positive controls and negative controls are diluted 1:30 with assay buffer then co-incubated with 250 ng/mL biotinylated-LT3114 and ruthenylated-LT3114 Master Mix on a polypropylene microtiter plate for 50-70 minutes. During this time, anti-LT3114 antibodies will bind to both the biotinylated-LT3114 and ruthenylated-LT3114 molecules to form an antibody complex bridge. After incubation, 50 µL of each mixture is added to the wells of the blocked Streptavidin coated MSD plate and allowed to bind for 1-1.5 hours. In the presence of tripropylamine-containing read buffer, ruthenium produces a chemiluminescent signal that is triggered when voltage is applied. The resulting chemiluminescence is measured in response units (RU).

Data Acquisition:

Equipment	MSD Sector Imager 6000
Program	Discovery Workbench™ 2006 MSD 3.0.17_3

10. RESULTS AND DISCUSSION

Table 1 lists all the validation runs performed.

10.1. Precision

Precision of the method was expressed as the coefficient of variation (%CV) from analyzing replicates of HPC, MPC and LPC. The mean RU of PCs and the mean RU of NCs in each run were evaluated. The assay will be considered precise if the %CV of HPC, MPC and LPC from each set of PC (intra-run) and all accepted runs (inter-run) are less than 25.0%.

The intra-run precision was determined from NC analyzed at N=5, duplicate readings and N≥5, duplicate readings of HPC (1:500), MPC (1:2000) and LPC (1:6000) in monkey serum on intra-run plate. The precision of intra run for HPC, MPC, LPC and NC are 4.3%, 4.4%, 1.5%, and 3.4% respectively (Table 2).

The inter-run precision of the assay was determined from plate controls, HPC, MPC, and LPC (N≥2, duplicate readings) and NC (N=3, duplicate readings) on each plate. The mean RU of NC and PCs from all accepted validation runs were included and a minimum of 6 validation runs over at least 3 days were evaluated.

The results for inter-assay precision for HPC, MPC, LPC and NC are 11.4%, 9.8%, 9.0%, and 5.4% respectively (Table 3). Both intra-run and inter-run precisions met the acceptance criteria.

10.2. Screening Cut Point and Confirmatory Cut Point Determination

The variability of antibody response for 34 monkey serum samples not exposed to LT3114 was evaluated to determine the assay cut point. The monkey serum samples were diluted at 1:30 (MRD) before being assayed. The cut point was determined from the instrument signal of the negative control and individual matrix responses, based on 95% upper confidence limit. The non-transformed data were evaluated for outliers, outliers were excluded and the data were assessed for normality. Two out of 3 non-transformed data sets displayed a non-normal distribution, therefore, all data were log transformed and re-evaluated. The log transformed data were evaluated for outliers, outliers were excluded and the data were assessed for normality. The details of the statistical evaluation were listed in Appendix C.

Screening data from each of the three days using log transformed data were found to be normally distributed using JMP statistical software. The daily cut point factor was calculated as follows:

*Daily Screening Cut Point = Log_{10} transformed mean signal from individual matrix samples + 1.645 * SD of Log_{10} transformed individual matrix samples*

The cut point factor (normalization factor, N) was calculated as follows:

N=Anti log (log transformed daily screening cut point – log transformed negative control of each individual data set)

The average calculated plate cut point factor for the screening runs of the monkey serum is 1.18 (Table 4). For each run, the floating cut point is determined by the normalization factor applied to biological background using the formula below:

Plate-specific Cut Point= 1.18 (N) × mean signal of negative control on each plate

The confirmatory cut point was validated to determine whether a positive response for ADA was specific to LT3114. Thirty four individual lots of monkey serum, as well as high and low PCs (HPC 1:500 and LPC 1:6000) were tested in the presence of 100 µg/mL LT3114, as well as no drug. For each sample, the percent of signal inhibition for each of the individual matrix samples was calculated. Outliers were identified by JMP statistical software. The details of the statistical evaluation were listed in Appendix C.

After outliers were excluded, the mean and SD of the percent inhibition was also determined and used to calculate confirmatory cut point as following:

Confirmatory cut point = Mean % signal inhibition of individual matrix samples

*+ 3.09*SD*

Calculated confirmatory cut point is 38.0% (Table 5A). This confirmatory assay cut point is accepted as the LT3114 spiked high and low PC (1:500 and 1:6000, respectively) samples run as controls had a % inhibition greater than the calculated confirmatory cut point. In the study, the % inhibition is 93.9% for HPC and 65.2% for LPC (Table 5B), which are both higher than the calculated confirmatory cut point 38.0%.

10.3. Matrix Selectivity (Recovery)

Matrix components can inhibit the ADA from binding to the capture and/or detection reagents under assay conditions. To evaluate matrix interference, 10 individual monkey serum lots, pooled monkey serum (NC) and buffer were spiked with HPC and LPC ($n \geq 1$) and analyzed at MRD 1:30 in the assay. Percent recovery was calculated for each individual monkey serum lot as:

%Recovery=(Mean Response of Spiked Lot) /(Mean Response of Spiked NC) x 100%

100% of the lots tested recovered within 75-125% when spiked with b HPC (1:500) and 90% of the lots tested recovered within 75-125% when spiked with LPC (1:6000) (Table 6).

10.4. LT3114 Drug Tolerance

In the drug tolerance test, the concentration of LT3114 that inhibits the ability to detect positive controls was assessed. High and low control PCs (HPC 1:500 / LPC 1:6000) and NC were incubated in the absence and presence of LT3114 at 200, 100, 50, 25, 10, 5, 0 µg/mL for ≥30 minutes prior to be evaluated in the assay.

The highest concentration of LT3114 that still appears positive or greater than the plate cut point for each positive control was identified. Results indicate that HPC (1:500 positive sera) can tolerate at least 100 µg/mL LT3114 (Table 7A). For the LPC, inconclusive results were obtained for the 5 µg/mL of LT3114, the lowest level of drug tested as one of two tests showed a positive response. Additional drug tolerance test was performed at lower drug levels, 1, 2, 3 and 4 µg/mL of LT3114 for the LPC (1:6000) and NC (Table 7B). The LPC (1:6000) can tolerate at least 4 µg/mL of LT3114 (Table 7B).

10.5. Stability

10.5.1. Freeze/Thaw Stability of Positive Control

Freeze/thaw stability of anti-LT3114 positive rabbit positive serum diluted in monkey serum was used as the surrogate to evaluate anti-LT3114 stability in the matrix after 5 freeze/thaw cycles. HPC(1:500), MPC (1:2000), and LPC (1:6000) (n=3) were frozen for at least 24 hours at -70°C and then subjected to 5 freeze/thaw cycles. Samples were allowed to thaw at room temperature under normal assay conditions for at least one hour and then frozen back to -70°C or colder for at least 12 hours. The treated PC as well as untreated PC (1 freeze/thaw cycle) was used to analyze in the assay to determine the FT stability. The results showed that all 3 levels of PC met the acceptance criteria with % Diff at -6.5%, -8.6% and -9.0% for HPC, MPC and LPC respectively (Table 9), suggesting that the positive control is stable for at least 5 freeze/thaw cycles at -70°C.

10.5.2. Room Temperature Stability of Positive Control

Room temperature stability of LT3114 anti-idiotype rabbit positive serum in rat serum was used as the surrogate to evaluate anti LT3114 stability in the matrix for the time that the samples were at room temperature during sample handling and processing. HPC (1:500), MPC (1:2000), and LPC (1:6000) (n=3) were thawed at room temperature under normal assay conditions and stored at room temperature for 24 hours. After incubation at room temperature, the PCs were frozen back at -70°C for at least 12 hours and then tested with the untreated PC (not exposed to room temperature as the treated PC). The results showed that all 3 levels of PC met the acceptance criteria with % Diff at -9.2%, -5.0% and -8.1% for HPC, MPC and LPC respectively (Table 8), suggesting that the positive control is stable for at least 24 hours at room temperature.

10.6. Sample Analysis Acceptance Criteria Determination

Based on the results from the validation, the following are the sample analysis acceptance criteria.

For a sample analysis run, HPC, MPC and LPC (N=2, duplicate readings) and NC (N=3, duplicate readings) will be included on each plate as controls.

Run acceptance criteria:

- For NC and PCs, the %CV from duplicate measurements should not be greater than 25% unless both measurements are below the plate cut point. Otherwise, the data is not reportable.
- At least two-thirds of NCs should be reportable.
- Plate NC Mean RU for each run must be ≤ 142 (Inter-assay Mean NC x 1.5).
- The mean values of the ECL responses of NC and three levels of PCs must be in the order of: NC< LPC< MPC <HPC.
- For the plate to be accepted at least 50% of the PCs at each level should be reportable and at least two-thirds of all PCs should be reportable.

Study samples at MRD (1:30) from each sample will be tested first by a screening assay. The %CV of RU from duplicate measurements should be ≤ 25% if at least one of the responses is ≥ the cut point on the same plate. Otherwise, the data is not reportable and the sample will be re-assayed.

(1) The plate cut point will be calculated using the plate mean NC * 1.18.

(2) If the mean RU of a sample is < plate cut point, the sample is reported negative.

(3) If the mean RU of a sample yields a mean RU ≥ plate cut point, the sample will be evaluated in the confirmation assay

(4) All screened positive samples will be analyzed both spiked and un-spiked with 100 µg/mL LT3114.

(5) For the confirmatory results to be accepted, at least 50% of the confirmatory PCs (HPC 1:500 and LPC 1:6000) at each level should be acceptable and at least two-thirds of all confirmatory PCs should be acceptable (i.e., reportable, with ≥38.0% %inhibition).

(6) Samples with % inhibition less than the validated CCP (38.0%) will be reported as "Negative" for the confirmatory result.

(7) Samples with an unspiked mean response equal to or greater than the plate cut point and % inhibition greater than or equal to the validated CCP (38.0%) will be reported as "Positive" for the confirmatory result.

(8) If the unspiked mean response is below the plate cut point with %inhibition ≥ 38.0%, the sample will be reassayed using the confirmation assay and the results of the reassay will be reported.

11. STUDY SPECIFIC DEVIATIONS OR OBSERVATIONS

There are no study specific deviations or observations that affect the data reported in the study.

12. CONCLUSION

An ECL method (QPS method 344-1404) has been validated for the detection of anti-LT3114 in monkey serum.

13. REFERENCES

QPS Validation Plan
QPS Validation Plan 344-1404

QPS SOPs
QPS-TLM-001: Validation of Immunogenicity Assays
GN-014-12A: Archives
DM-010-01 Immune Response Module Using Watson LIMS

Study Notebooks
Validation Notebook QPS 344-1404
Client 344 Compound Receipt Notebook
Client 344 LT3114 Assay Notes
Client 344 Reagent Receipt Notebook

White Paper
Shankar G., Devanarayan V., Amaravadi L., et al. Recommendations for the Validation of Immunoassays Used for Detection of Host Antibodies Against Biotechnology Products. J. Pharm. Biomed. Anal. 2008, 48, 1267-1281.

14. ARCHIVE STATEMENT

All raw data (including electronic SoftmaxPro files), a copy of this bioanalytical report and required supporting information for this study will be held under the control of QPS, LLC, 3 Innovation Way, Suite 240, Newark DE, 19711, USA. Study records will be archived according to the most current version of SOP GN-014, "Archives".

15. TABLES

Table 1 Validation Run Log

Run Number	Run ID	Validaton Items	Monkey Serum Lots Tested	Run Status[a]
1	140716-3441404-MKSE-RBU-1A-001	Intra-Day, Room Temperature Stability	NA	Accepted
2	140716-3441404-MKSE-RBU-2B-002	Screening (Day 1)	MKSE 1 - 17 and 21 - 37	Accepted
3	140716-3441404-MKSE-QMA-1A-003	LT314 Confirmation Plate 1	MKSE 1 - 8 and 29 - 37	Accepted[c]
4	140716-3441404-MKSE-QMA-2B-004	LT314 Confirmation Plate 2	MKSE 9-17 and 21 - 28	Accepted[c]
5	140717-3441404-MKSE-QMA-1A-005	Screening (Day 2)	MKSE 1-17 and 21-37	Accepted
6	140717-3441404-MKSE-QMA-2B-006	Matrix Selectivity (Recovery)	MKSE 1-5 and 21-25	Accepted
7	140718-3441404-MKSE-TC-1A-007	Screening (Day 1)	MKSE 1-17 and 21-37	Accepted
8	140718-3441404-MKSE-TC-2B-008	Freeze-Thaw Stability and Drug Tolerance	NA	Accepted[b]
9	140718-3441404-MKSE-TC-3C-009	Drug Tolerance	NA	Accepted[b]
10	140721-3441404-MKSE-RBU-1A-010	Drug Tolerance	NA	Accepted
11	140721-3441404-MKSE-RBU-2B-011	Drug Tolerance	NA	Accepted
12	140724-3441404-MKSE-BLB-1A-012	Drug Tolerance (LPC and NC)	NA	Accepted

a: Run accepted / rejected according to QPS VA protocol 344-1404

b: Plate passes run criteria. Preparation error occurred during drug tolerance test, therefore data are not reportable.

c: Due to planning error for Runs 3 and 4, data were processed in Watson Runs 16 and 17, respectively

Table 2 Intra-Run Precision of the Anti-LT3114 Positive Rabbit Serum for the Detection Assay in Monkey Serum

Replicate	HPC (1:500)		MPC (1:2000)		LPC (1:6000)		Negative Control	
	Mean	%CV	Mean	%CV	Mean	%CV	Mean	%CV
1	1579	1.5	459.5	3.2	220.0	1.3	93.00	4.6
2	1769	1.2	476.0	4.8	223.0	1.9	91.50	2.3
3	1650	2.0	483.0	1.8	226.5	2.2	96.00	1.5
4	1642	2.1	516.0	1.4	229.0	4.3	93.50	5.3
5	1702	3.6	495.0	1.1	224.5	0.9	87.50	0.8
Mean	1668		485.9		224.6		92.30	
SD	71.199		21.167		3.417		3.134	
%CV	4.3		4.4		1.5		3.4	

Run ID: 140716-3441404-MKSE-RBU-1A-001

Table 3 Inter-Run Precision of the Anti-LT3114 Positive Rabbit Serum for the Detection Assay in Monkey Serum

Run Date	Run Number	HPC (1:500)		MPC (1:2000)		LPC (1:6000)		Negative Control	
		Average	CV%	Average	CV%	Average	CV%	Average	CV%
7/16/2014	1	1579	1.5	459.5	3.2	220.0	1.3	93.0	4.6
		1769	1.2	476.0	4.8	223.0	1.9	91.5	2.3
		1650	2.0	483.0	1.8	226.5	2.2	96.0	1.5
		1642	2.1	516.0	1.4	229.0	4.3	93.5	5.3
		1702	3.6	495.0	1.1	224.5	0.9	87.5	0.8
7/16/2014	2	1450	2.6	480.5	0.1	228.5	4.6	99.5	3.6
		1543	4.9	483.0	3.5	228.5	0.3	99.0	1.4
		NA	NA	NA	NA	NA	NA	96.0	2.9
7/16/2014	3	1596	1.2	462.5	0.2	216.5	2.3	87.5	5.7
		1681	2.1	460.0	0.0	213.5	3.0	89.0	0.0
		NA	NA	NA	NA	NA	NA	89.0	0.0
7/16/2014	4	1637	0.6	479.0	1.5	224.5	0.3	95.5	5.2
		1758	0.3	475.5	0.4	228.0	0.0	95.0	0.0
		NA	NA	NA	NA	NA	NA	95.5	3.7
7/17/2014	5	1724	2.4	513.0	1.4	236.0	0.6	93.5	2.3
		1872	0.1	539.0	1.0	248.5	0.9	88.5	2.4
		NA	NA	NA	NA	NA	NA	92.0	3.1
7/17/2014	6	1739	2.7	506.5	1.8	234.5	0.9	100	1.4
		1880	0.3	521.0	0.8	246.5	0.9	96.5	0.7
		NA	NA	NA	NA	NA	NA	96.0	4.4
7/18/2014	7	2274	3.0	642.5	2.1	286.5	2.7	105	0.0
		2059	2.3	588.0	3.8	273.0	1.6	106	2.0
		NA	NA	NA	NA	NA	NA	102	1.4
7/18/2014	8	1611	2.4	492.0	1.7	225.0	1.9	89.0	3.2
		1466	1.4	430.0	1.6	194.0	0.0	98.0	8.7
		1845	2.9	554.0	3.6	256.0	2.8	93.5	2.3
7/18/2014	9	1283	2.7	391.0	0.4	185.0	1.5	85.5	0.8
		1590	0.9	468.0	0.3	216.0	2.0	85.5	0.8
		NA	NA	NA	NA	NA	NA	88.0	3.2
7/21/2014	10	1699	3.0	491.0	1.2	223.0	3.8	95.5	6.7
		1713	2.5	491.5	0.7	227.5	2.2	96.0	1.5
		NA	NA	NA	NA	NA	NA	97.5	2.2
7/21/2014	11	1600	0.0	479.0	1.8	223.5	2.8	98.5	2.2
		1619	1.4	485.0	0.6	222.0	1.3	97.0	5.8
		NA	NA	NA	NA	NA	NA	97.0	1.5
7/24/2014	12	1502	2.4	483.5	0.4	228.0	1.2	116	6.7
		1518	0.8	482.0	2.1	235.5	0.3	103	3.4
		NA	NA	NA	NA	NA	NA	112	0.6
	Mean	1692		494.7		229.2		94.5	
	SD	193.695		48.4		20.7		5.116	
	%CV	11.4		9.8		9.0		5.4	

Table 4 **Screening Assay Cut Point Evaluation of Individual Monkey Serum Samples in the Anti-LT3114 Detection Assay**

Designated Lot#	Bioreclamation Lot#	Gender	Day 1			Day 2			Day 3		
			RU[a]	Log_{10} (RU)	CV%	RU[b]	Log_{10} (RU)	CV%	RU[c]	Log_{10} (RU)	CV %
MKSE 1	CYN141417	Male	106.5	2.027	2.0	103.5	2.015	2.0	109.5	2.039	1.9
MKSE 2	CYN141418	Male	109.0	2.037	2.6	105.0	2.021	1.3	109.0	2.037	1.3
MKSE 3	CYN141419	Male	99.00	1.996	0.0	96.00	1.982	2.9	98.00	1.991	0.0
MKSE 4	CYN141420	Male	102.5	2.011	2.1	97.50	1.989	0.7	95.00	1.978	1.5
MKSE 5	CYN141421	Male	103.5	2.015	0.7	100.5	2.002	0.7	99.00	1.996	2.9
MKSE 6	CYN141422	Male	111.0	2.045	1.3	107.5	2.031	0.7	105.5	2.023	0.7
MKSE 7	CYN141423	Male	109.5	2.039	0.6	107.0	2.029	1.3	104.0	2.017	0.0
MKSE 8	CYN141424	Male	104.0	2.017	0.0	164.0[g]	NR	32.8	96.50	1.985	0.7
MKSE 9	CYN141425	Male	102.0	2.009	4.2	100.0	2.000	4.2	99.50	1.998	5.0
MKSE 10	CYN141426	Male	106.5	2.027	0.7	102.0	2.009	2.8	110.5	2.043	0.6
MKSE 11	CYN141427	Male	125.5	2.099[d]	3.9	124.0	2.093	2.3	134.0	2.127[d]	0.0
MKSE 12	CYN141428	Male	103.5	2.015	2.0	104.5	2.019	3.4	100.5	2.002	3.5
MKSE 13	CYN141429	Male	107.0	2.029	1.3	110.0	2.041	1.3	104.0	2.017	0.0
MKSE 14	CYN141430	Male	117.5	2.070	3.0	122.0	2.086	1.2	121.5	2.085	1.7
MKSE 15	CYN141431	Male	98.50	1.993	0.7	96.00	1.982	2.9	97.00	1.987	2.9
MKSE 16	CYN141432	Male	95.50	1.980	2.2	97.50	1.989	0.7	93.00	1.968	1.5
MKSE 17	CYN141433	Male	96.50	1.985	0.7	94.50	1.975	3.7	102.0	2.009	1.4
MKSE 21	CYN141435	Female	97.00	1.987	0.0	91.50	1.961	5.4	92.50	1.966	0.8
MKSE 22	CYN141436	Female	98.50	1.993	3.6	89.00	1.949	0.0	91.00	1.959	0.0
MKSE 23	CYN141437	Female	105.5	2.023	2.0	99.50	1.998	2.1	102.5	2.011	0.7
MKSE 24	CYN141438	Female	96.50	1.985	2.2	91.50	1.961	2.3	93.00	1.968	3.0
MKSE 25	CYN141439	Female	102.5	2.011	0.7	110.0	2.041	0.0	103.5	2.015	4.8
MKSE 26	CYN141440	Female	96.00	1.982	4.4	97.00	1.987	2.9	92.00	1.964	0.0
MKSE 27	CYN141441	Female	92.50	1.966	2.3	97.00	1.987	1.5	93.00	1.968	1.5
MKSE 28	CYN141442	Female	94.00	1.973	3.0	100.0	2.000	2.8	87.00	1.940	1.6
MKSE 29	CYN141443	Female	105.5	2.023	6.0	105.0	2.021	5.4	98.50	1.993	0.7
MKSE 30	CYN141444	Female	165.5[g]	NR	56.0	92.00	1.964	0.0	93.00	1.968	3.0
MKSE 31	CYN141445	Female	100.0	2.000	4.2	92.00	1.964	3.1	90.50	1.957	2.3
MKSE 32	CYN141446	Female	100.5	2.002	0.7	99.00	1.996	4.3	91.50	1.961	2.3
MKSE 33	CYN141447	Female	91.00	1.959	1.6	94.50	1.975	2.2	89.00	1.949	0.0
MKSE 34	CYN141448	Female	103.0	2.013	1.4	111.0	2.045	1.3	111.5	2.047	0.6
MKSE 35	CYN141449	Female	87.50	1.942	2.4	95.50	1.980	0.7	89.50	1.952	2.4
MKSE 36	CYN141450	Female	106.5	2.027	0.7	118.5	2.074	1.8	104.5	2.019	4.7
MKSE 37	CYN141451	Female	112.0	2.049	2.5	111.0	2.045	0.0	121.5	2.085	4.1
N				32			33			33	
Mean Log_{10}(RU) of Individual Monkey Serum				2.007			2.006			1.997	
SD				0.028			0.036			0.04	
%CV				1.4			1.8			2.0	
Daily Screening Cut Point[e]				2.053			2.065			2.063	
Log_{10}(Mean NC)				1.992			1.960			2.017	
Normalization Factor[f]				1.15			1.27			1.11	
Average Cut Point Factor							**1.18**				

Table 4 (Cont.) Screening Assay Cut Point Evaluation of Individual Monkey Serum Samples in the Anti-LT3114 Detection Assay

RU: Response Unit

a: Run ID: 140716-3441404-MKSE-RBU-2B-002

b: Run ID: 140717-3441404-MKSE-QMA-1A-005

c: Run ID: 140718-3441404-MKSE-TC-1A-007

d: Statistical outliers excluded from the screening cut point assessement by Box-Plot using JMP statistical software

e: Daily Screening Cut Point: Mean response of individual Serum from one day of Analysis + (1.645 × SD of individual serum from one day of analysis)

f: Normalization Factor = Anti Log (Log transformed daily screening cut point - log transformed negative control for each individual data set) (QPS protocol 344-1404)

g: Not reportable due to %CV >25%

Table 5A Confirmatory Assay Cut Point of Anti-LT3114 Antibody Assay in Monkey Serum

Lot ID	Lot Number	Gender	Run No.	NS	%CV	LT3114[a]	%CV	% Inhibition[b]
MKSE 1	CYN141417	Male	3	96.0	1.5	71.0	10.0	26.04
MKSE 2	CYN141418	Male	3	102.5	2.1	70.5	3.0	31.22
MKSE 3	CYN141419	Male	3	92.5	0.8	68.5	5.2	25.95
MKSE 4	CYN141420	Male	3	92.5	2.3	72.0	3.9	22.16
MKSE 5	CYN141421	Male	3	87.0	1.6	63.5	3.3	27.01
MKSE 6	CYN141422	Male	3	91.5	3.9	63.0	0.0	31.20
MKSE 7	CYN141423	Male	3	87.0	1.6	67.0	6.3	22.99
MKSE 8	CYN141424	Male	3	84.5	0.8	61.5	3.4	27.22
MKSE 9	CYN141425	Male	4	102.0	2.8	77.5	6.4	24.02
MKSE 10	CYN141426	Male	4	100.0	2.8	76.5	0.9	23.50
MKSE 11	CYN141427	Male	4	113.0	1.3	78.5	0.9	30.53
MKSE 12	CYN141428	Male	4	106.0	1.3	77.0	5.5	27.36
MKSE 13	CYN141429	Male	4	110.0	1.3	117.5	41.5	NR
MKSE 14	CYN141430	Male	4	114.0	0.0	88.0	1.6	22.81
MKSE 15	CYN141431	Male	4	101.5	0.7	79.0	1.8	22.17
MKSE 16	CYN141432	Male	4	102.0	1.4	80.5	0.9	21.08
MKSE 17	CYN141433	Male	4	110.0	2.6	76.0	3.7	30.91
MKSE 21	CYN141435	Female	4	97.0	1.5	78.0	1.8	19.59
MKSE 22	CYN141436	Female	4	94.5	3.7	72.0	2.0	23.81
MKSE 23	CYN141437	Female	4	95.5	3.7	76.0	1.9	20.42
MKSE 24	CYN141438	Female	4	92.0	1.5	72.0	2.0	21.74
MKSE 25	CYN141439	Female	4	108.0	3.9	83.5	0.8	22.69
MKSE 26	CYN141440	Female	4	99.5	0.7	77.5	0.9	22.11
MKSE 27	CYN141441	Female	4	94.0	1.5	79.0	1.8	15.96
MKSE 28	CYN141442	Female	4	99.5	0.7	84.0	5.1	15.58
MKSE 29	CYN141443	Female	3	94.0	1.5	72.5	1.0	22.87
MKSE 30	CYN141444	Female	3	96.5	0.7	72.0	0.0	25.39
MKSE 31	CYN141445	Female	3	92.0	4.6	80.5	14.9	12.50[c]
MKSE 32	CYN141446	Female	3	91.5	0.8	72.0	2.0	21.31
MKSE 33	CYN141447	Female	3	83.0	1.7	66.0	0.0	20.48
MKSE 34	CYN141448	Female	3	100.0	0.0	64.5	1.1	35.50
MKSE 35	CYN141449	Female	3	81.5	0.9	64.0	0.0	21.47
MKSE 36	CYN141450	Female	3	102.0	0.0	74.5	2.8	26.96
MKSE 37	CYN141451	Female	3	97.0	2.9	70.5	1.0	27.32
							N	32
							Mean % Inhibition	24.36
							SD	4.42

Table 5A (Cont.) Confirmatory Assay Cut Point of Anti-LT3114 Antibody Assay in Monkey Serum

Mean % Inhibition + (3.09*SD)	38.0

Confirmation Runs: 140717-3441404-MKSE-QMA-1A-003 and 140717-3441404-MKSE-QMA-2B-004

a: Samples were incubated in the presence of 100 µg/mL LT3114 before being processed.

b: 100 x (1 - RU of LT3114 Spiked Sample/RU of Unspiked Sample)
c: Data excluded from confirmatory cut point calculation due to lot being identified as outlier by Box-Plot Analysis (QPS Protocol 344-1404)

NS: Non-spiked (neat)

Table 5B Confirmation Inhibition of Anti-LT3114 Positive Rabbit Serum in Monkey Serum

Run Number	HPC (1:500)					LPC (1:6000)				
	NS	%CV	Spiked [b]	%CV	%Inhibition[a]	NS	%CV	Spiked[b]	%CV	%Inhibition[a]
3	1596	1.2	97.00	1.5	93.9	216.5	2.3	72.00	2.0	66.7
	1681	2.1	97.00	1.5	94.2	213.5	3.0	70.50	1.0	67.0
4	1637	0.6	108.5	2.0	93.4	224.5	0.3	84.00	0.0	62.6
	1758	0.3	107.5	0.7	93.9	228.0	0.0	81.00	3.5	64.5
N	4		4		4	4		4		4
Mean	1668		102.5		93.9	220.6		76.88		65.2
SD	69		6.36		0.33	6.8		6.64		2.06
%CV	4.1		6.2		0.4	3.1		8.6		3.2

Confirmation Runs: 140716-3441404-MKSE-QMA-1A-003 and 140716-3441404-MKSE-QMA-2B-004

a: %Inhibition = 100*(1 - LT3114 spiked RU/ NS RU)

b: Spiked: Samples were incubated in the presence of 100 µg/mL LT3114 before being processed.

NS: Non-spiked sample

Table 6 Matrix Selectivity (Recovery) Test of Anti-LT3114 Positive Rabbit Serum in Monkey Serum

Designated Lot#	Bioreclamation/ QPS Lot Number	Unspiked RU	Spiked with HPC (1:500) RU	% Recovery	Spiked with LPC (1:6000) RU	% Recovery
Pooled MKSE	TC120102-01	99.00[a]	1730[a]	100.0	238.8[a]	100.0
Buffer	NA	103.5	2633	152.2	309.5	129.6
MKSE 01	CYN141417	104.5	1852	107.1	248.5	104.1
MKSE 02	CYN141418	104.5	1990	115.0	259.5	108.7
MKSE 03	CYN141419	91.50	1886	109.0	326.5	136.7
MKSE 04	CYN141420	94.00	1944	112.4	239.0	100.1
MKSE 05	CYN141421	99.00	2040	117.9	277.5	116.2
MKSE 21	CYN141435	93.50	2117	122.4	275.0	115.2
MKSE 22	CYN141436	91.00	1860	107.5	256.5	107.4
MKSE 23	CYN141437	97.50	1953	112.9	262.5	109.9
MKSE 24	CYN141438	92.50	2089	120.8	247.0	103.4
MKSE 25	CYN141439	104.0	1947	112.5	253.0	105.9

RU: Response Unit
Run ID: 140717-3441404-MKSE-QMA-2B-006
% Recovery = (Mean Spiked Lot RU / Mean Spiked Pooled MKSE RU) x 100%
a: Sample analyzed at N=2. Mean of replicates is reported for selectivity evaluation.

Table 7A LT3114 Drug Tolerance (5-200 µg/mL of LT3114) in the Anti-LT3114 Antibody Detection Assay in Monkey Serum Using the Anti-LT3114 Positive Rabbit Serum (HPC, LPC and NC)

LT3114 (µg/mL)	Response Units (RU)					
	Test 1: Run 010			Test 2: Run 011		
	HPC 01 (1:500)	LPC 01 (1:6000)	NC 01	HPC 02 (1:500)	LPC 02 (1:6000)	NC 02
0	1588	205.5	90.00	1557	206.0	92.50
5	424.5	**115.0**	84.50	435.5	114.5	86.50
10	316.5	106.0	84.50	317.0	107.5	84.50
25	210.5	97.50	88.00	214.0	93.50	84.50
50	156.0	88.50	88.00	159.0	86.50	83.50
100	**117.5**	84.00	82.00	**120.5**	79.50	82.00
200	88.00	76.50	75.00	89.00	79.00	72.50
Mean RU of Plate NC	96.3			97.5		
Plate Cut Point[a]	114			115		

a: Plate Cut Point: Mean RU of Plate NC * Cut Point Factor (1.18) as determined by Watson 7.4.1

Run IDs: 140721-3441404-MKSE-RBU-1A-010 and 140721-3441404-MKSE-RBU-2B-011

Highest level of LT3114 where PC levels are ≥ plate cut point are bolded

Table 7B LT3114 Drug Tolerance (1-4 µg/mL of LT3114) in the Anti-LT3114 Antibody Detection Assay in Monkey Serum Using the Anti-LT3114 Positive Rabbit Serum (LPC and NC)

LT3114 (µg/mL)	**Relative Units (RU)**			
	Test 1 and 2: Run 012			
	LPC 01 (1:6000)	**LPC 02 (1:6000)**	**NC 01**	**NC 02**
0	231.0	224.5	115.5	109.5
1	174.0	169.5	112.5	114.5
2	165.0	159.5	118.0	104.5
3	155.0	149.5	116.0	112.5
4	**141.0**	**142.5**	106.0	114.0
Mean RU of Plate NC	110			
Plate Cut Point[a]	130			

a: Plate Cut Point: Mean RU of Plate NC * Cut Point Factor (1.18) as determined by Watson 7.4.1

Run IDs: 140724-3441404-MKSE-BLB-1A-012

Highest level of LT3114 where PC levels are ≥ plate cut point are bolded

Table 8 Room Temperature Stability of Anti-LT3114 Positive Rabbit Serum in Monkey Serum

Positive Controls	Untreated RU values				Treated (24hr RT) RU values				% Difference
	Mean	%CV			Mean	%CV			
HPC 01	1579	1.5	Mean	1668	1518	2.6	Mean	1515	-9.2
HPC 02	1769	1.2	SD	71.199	1512	4.6	SD	3.055	
HPC 03	1650	2.0	%CV	4.3	1516	2.8	%CV	0.2	
HPC 04	1642	2.1			NA	NA			
HPC 05	1702	3.6			NA	NA			
MPC 01	459.5	3.2	Mean	485.9	458.5	2.9	Mean	461.8	-5.0
MPC 02	476.0	4.8	SD	21.167	464.0	0.3	SD	2.930	
MPC 03	483.0	1.8	%CV	4.4	463.0	1.8	%CV	0.6	
MPC 04	516.0	1.4			NA	NA			
MPC 05	495.0	1.1			NA	NA			
LPC 01	220.0	1.3	Mean	224.6	206.0	0.0	Mean	206.5	-8.1
LPC 02	223.0	1.9	SD	3.417	206.0	1.4	SD	0.866	
LPC 03	226.5	2.2	%CV	1.5	207.5	1.0	%CV	0.4	
LPC 04	229.0	4.3			NA	NA			
LPC 05	224.5	0.9			NA	NA			

PC: Anti-LT3114 Positive Rabbit Serum in pooled Monkey Serum (HPC 1:500, LPC 1:2000 & LPC 1:6000)
Run ID: 140716-3441404-MKSE-RBU-1A-001
% Difference: 100*((Mean 24 hour treated PC-Mean untreated PC)/(Mean untreated PC))
RU: Response Unit

Table 9 Freeze / Thaw Stability of Anti-LT3114 Positive Rabbit Serum in Monkey Serum

PC Level	Untreated RU values				Treated (5 F/T Cycles) RU values				% Difference
	Mean	%CV			Mean	%CV			
HPC 01	1611	2.4	Mean	1641	1229	0.9	Mean	1534	-6.5
HPC 02	1466	1.4	SD	191.234	1743	1.9	SD	270.291	
HPC 03	1845	2.9	%CV	11.7	1631	0.8	%CV	17.6	
MPC 01	492.0	1.7	Mean	492.0	420.0	1.0	Mean	449.8	-8.6
MPC 02	430.0	1.6	SD	62.000	456.0	2.2	SD	27.278	
MPC 03	554.0	3.6	%CV	12.6	473.5	0.7	%CV	6.1	
LPC 01	225.0	1.9	Mean	225.0	184.5	1.9	Mean	204.8	-9.0
LPC 02	194.0	0.0	SD	31.000	198.0	1.4	SD	24.476	
LPC 03	256.0	2.8	%CV	13.8	232.0	2.4	%CV	12.0	

PC: Anti-LT3114 Positive Rabbit Serum in pooled Monkey Serum (HPC 1:500, MPC 1:2000 & LPC 1:6000)
Run ID: 140718-3441404-MKSE-TC-2B-008
%Difference: 100*((Mean F/T hour treated PC-Mean untreated PC)/(Mean untreated PC))
RU: Response Unit

16. APPENDICES

Appendix A: Certificate of Analysis for LT3114

Lpath Incorporated	Certificate of Analysis Form				
TITLE	**Certificate of Analysis for Monoclonal Antibody Produced by Lpath**				**LOT NUMBER** LP121416
DOCUMENT NUMBER	**REVISION**	**SUPERCEDED REVISION**	**ISSUE DATE**	**EFFECTIVE DATE**	**PAGE**
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	NOV 1, 2013	1 of 5

Name	Part Number	Lot Number
Humanized Lpathomab	LT3114	LP121416

Formulation Buffer	Storage Temperature
PBS (Mediatech #21-040, Lot# 21040283)	2 to 8 °C

Production Date	Retest Date	Project /Use
5-10-2013	N/A	TBD

Summary

Production location, date, and method
The antibody was produced from stable CHO clone (191+192) 3114 3B8/1B4/2F1 in 4x 3-L shake flasks at Lpath. Takedown on d10, May 10, 2013. Pooled, clarified harvest was stored at -20 °C.

Lot LP121416 consists of 3x 1-L bottles of supe which cracked during storage.
Lot LP121417 consists of 1x 1-L bottle of supe which did not crack during storage.
Purified lots LP121416 and LP121417 were pooled to create lot LP121419.

Purification location, date and method
Antibody purification was initiated on August 21, 2013 at Lpath using affinity and AEX chromatography (MabSelect Protein A and Q membrane Sartorius #92IEXQ42D4-OOA, respectively). Following concentration, the antibody was dialyzed against 3x 2-L PBS (Mediatech) and passed through a 0.22 um filter flask unit (Corning).

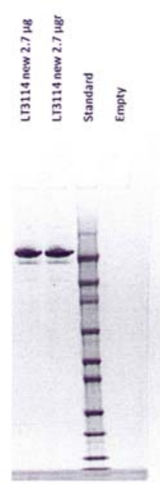
Concentration by A_{280} and extinction coefficient
The final sample is 21.5 mg/mL as determined by UV spectroscopy using a theoretical extinction coefficient of 1.4 mL/mg.

Total amount of Ab (mg)
1400 mg

Appearance details:
Clear, slightly yellow solution

Confidential

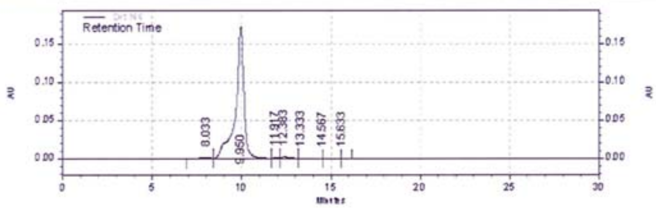
Lpath INCORPORATED	Certificate of Analysis Form				
TITLE	**Certificate of Analysis for Monoclonal Antibody Produced by Lpath**				**LOT NUMBER** **LP121416**
DOCUMENT NUMBER	**REVISION**	**SUPERCEDED REVISION**	**ISSUE DATE**	**EFFECTIVE DATE**	**PAGE**
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	NOV 1, 2013	2 of 5

Attribute	Test	Specifications	Results
Identity	Appearance	Particulate free, clear solution Colorless or slightly yellow	Particulate free, clear solution Slightly yellow
	SDS-PAGE (Non-Reduced)	Report Results	Major band at 165 kDa
	SDS-PAGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Two pronounced bands; One at ~55 kDa; One at ~ 28 kDa
	Insert images of SDS-PAGE	Non-Reduced 	Reduced 
	Isoelectric Focusing (by IEF Gel)	Report Results	Not done

Confidential

Lpath INCORPORATED	Certificate of Analysis Form				
TITLE	**Certificate of Analysis for Monoclonal Antibody Produced by Lpath**				**LOT NUMBER**
					LP121416
DOCUMENT NUMBER	**REVISION**	**SUPERCEDED REVISION**	**ISSUE DATE**	**EFFECTIVE DATE**	**PAGE**
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	NOV 1, 2013	3 of 5

Attribute	Test		Specifications	Results
Potency	Protein Concentration (OD-280)		Report Results	20.14 mg/mL
	LPA Binding Direct ELISA (EC 50)		Report Results	25.05 ng/mL
	Bioactivity Assay (IL-6 and EDG2)		Positive Result	Does not stimulate IL-6 EDG2 IC50 = 13.65 µg/mL
Purity	*Insert image of SE-HPLC*			



Det 166 Results

Time	Area	Area %	Height	Height %
8.033	68097	1.14	1413	0.78
9.950	5707101	95.95	173377	96.24
11.917	35431	0.60	1744	0.97
12.383	97022	1.63	2645	1.47
13.333	31081	0.52	668	0.37
14.567	8116	0.14	234	0.13
15.633	1015	0.02	62	0.03
Totals	5947863	100.00	180143	100.00

Attribute	Test		Specifications	Results
Purity	**SE-HPLC**	Purity [% as IgG monomer]	≥ 90%	96%
		Aggregates [as % of total peak area]	≤ 6%	1%
		Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%

Confidential

Lpath INCORPORATED	Certificate of Analysis Form				
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER: LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	NOV 1, 2013	4 of 5

Attribute	Test			Specifications	Results
	SDS-PAGE	reduced	Purity [%]	≥ 90%	≥ 90%
			50 kDa Band	~65%	~67%
			25 kDa Band	~35%	~33%
			Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%
	Endotoxin			≤ 10.0 EU/mg of protein	0.06 EU/mg
	Bioburden			≤ 10 CFU/mL	Not done

Comments

Source Data:

	Lpath notebook	Pages
Production	277	149
Purification	296	11,18,25
UV A280	296	48
SDS-PAGE	296	26
LPA binding ELISA	221	21-22
IL-6	237	48-49
EDG2	237	51
SE-HPLC	296	33-34
Endotoxin	298	13

Additional Comments on Sample Quality:

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Lpath INCORPORATED	Certificate of Analysis Form				
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER: LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	5 of 5

JUDGMENT
Approved for testing in humans ☐ - Accepted ☐ - Rejected **For Research Only** **Not For Use in Humans** Prepared by: ________ Date: 12-6 2013 Cindy Dickerson, Ph.D. Reviewed by: ________ Date: 12/6/ 2013 Jon Wojciak, Ph.D. Approved by: ________ Date: 3rd Decemb 2013 Gary Woodnutt, Ph.D.

Confidential

Appendix B: Anti-LT3114 Positive Rabbit Serum

Lpath
INCORPORATED

May 20, 2014

Rupalie Meegalla
QPS, LLC
1 Innovation Way, Suite 200
Newark, DE19711

Dear Rupalie,

Enclosed, please find the following materials in this shipment:

> Whole rabbit serum from bleed 2 of rabbit immunized with whole 3114 IgG aliquoted in 2 vials 1ml/each (Whole 3114 anti-idiotype)

Upon arrival, please stored at 4° C. Material stored at 4° C under sterile condition is stable for up to 2 months.
Should you have any questions regarding the material, please do not hesitate to contact us.

Best regards,

Rosalia Matteo, PhD
Asst. Director
Lpath Incorporated

Lot No: R-2265-B2-11/13 ①
Exp. Date: 5/21/16 ② BCB 7/25/14

① Refer to client email from 7/2/14 for clarification BCB 7/25/14
② Assigned per QPS SOP LP-006 (2 years from date of receipt. BCB 7/25/14

4025 Sorrento Valley Blvd., San Diego, CA 92121-1404 • Main: (858) 678-0800 • Fax: (858) 678-0900
www.Lpath.com

Appendix C Statistical Analysis for Screening Assay Cut Point and Confirmatory Cut Point Outlier Evaluation

All statistical analyses were completed using JMP Statistical Discovery Software (Version 10.0; SAS Institute, Inc., Cary, NC, USA). Statistical methods used for the analysis are consistent with procedures recommended by Shankar G, et al. 2008.

The average RU values and $\log_{10}$ transformed RU values of individual samples were imported into JMP. Readings that had %CV greater than 25% were excluded.

First, statistical outliers for screening using the non-transformed RU values were evaluated on a per-day basis using the "box-plot" method in JMP software. The detailed formula is also listed in QPS SOP QPS-TLM-001. Individuals that have a response above the upper boundary or below the lower boundary were exported as an Excel file and excluded from further analysis. Each exported Excel file is included for verification of outliers.

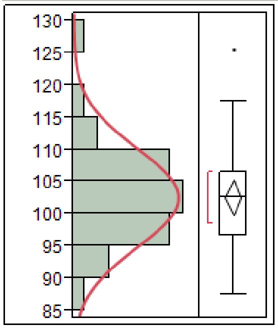
Next, the normality for screening analysis was assessed on a per-day basis using Shapiro-Wilk test in JMP software. The normality was indicated by the probability in the "Goodness-of-Fit". If the probability is greater and equal to 0.05, the data is normally distributed. If the probability is less than 0.05, the data is not normally distributed.

Day 1 (Screening Cut Point of Non-transformed Values):

Initial:

Distributions

Mean Response Without Drug



Normal(102.606,7.65645)

Quantiles

100.0%	maximum	125.5
99.5%		125.5
97.5%		125.5
90.0%		111.6
75.0%	quartile	106.5
50.0%	median	102.5
25.0%	quartile	96.75
10.0%		93.1
2.5%		87.5
0.5%		87.5
0.0%	minimum	87.5

Summary Statistics

Mean	102.60606
Std Dev	7.6564491
Std Err Mean	1.3328167
Upper 95% Mean	105.32092
Lower 95% Mean	99.891202
N	33

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	102.60606	99.891202	105.32092
Dispersion	σ	7.6564491	6.1572334	10.127132

-2log(Likelihood) = 226.996131409753

Goodness-of-Fit Test

Shapiro-Wilk W Test

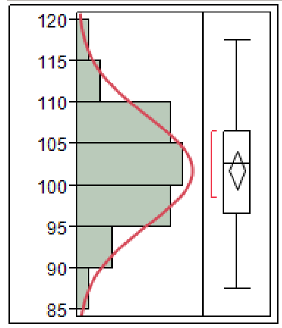
W	Prob<W
0.967525	0.4151

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

Final:

Distributions

Mean Response Without Drug



Normal(101.891,6.56326)

Quantiles

100.0%	maximum	117.5
99.5%		117.5
97.5%		117.5
90.0%		110.55
75.0%	quartile	106.5
50.0%	median	102.5
25.0%	quartile	96.625
10.0%		92.95
2.5%		87.5
0.5%		87.5
0.0%	minimum	87.5

Summary Statistics

Mean	101.89063
Std Dev	6.5632584
Std Err Mean	1.1602311
Upper 95% Mean	104.25693
Lower 95% Mean	99.524318
N	32

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	101.89063	99.524318	104.25693
Dispersion	σ	6.5632584	5.2617864	8.725718

-2log(Likelihood) = 210.227246136211

Goodness-of-Fit Test

Shapiro-Wilk W Test

W	Prob<W
0.993979	0.9996

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

The following individual MKSE lot was identified as an outlier:

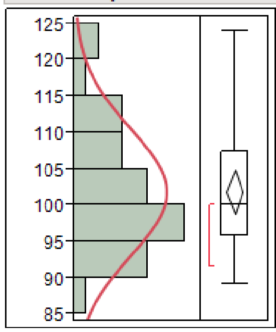
Sample Number	Lot Number	Assay Date	Mean Response Without Drug	CV Response Without Drug (%)	Log Mean Response Without Drug
11	CYN141427	16Jul2014	125.5	3.9	2.0986

Data distribution was assessed using JMP Statistical Discovery with the assumption that the data is normally distributed. The model of assumption was confirmed by Shapiro-Wilk W Test (P value = 0.9996). *The data are normally distributed.*

Day 2 (Screening Cut Point of Non-transformed Values):

Distributions

Mean Response Without Drug



Normal(101.864,8.76719)

Quantiles

100.0%	maximum	124
99.5%		124
97.5%		124
90.0%		115.5
75.0%	quartile	107.25
50.0%	median	100
25.0%	quartile	95.75
10.0%		91.7
2.5%		89
0.5%		89
0.0%	minimum	89

Summary Statistics

Mean	101.86364
Std Dev	8.7671909
Std Err Mean	1.526172
Upper 95% Mean	104.97235
Lower 95% Mean	98.754926
N	33

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	101.86364	98.754926	104.97235
Dispersion	σ	8.7671909	7.05048	11.596302

-2log(Likelihood) = 235.937028744337

Goodness-of-Fit Test

Shapiro-Wilk W Test

W	Prob<W
0.929484	0.0338*

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

Outliers were not present in the data from Day 2 of analysis.

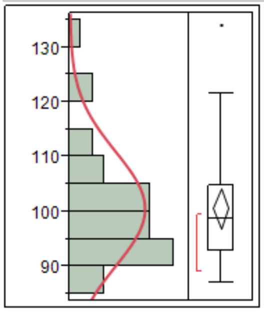
Data distribution was assessed using JMP Statistical Discovery with the assumption that the data is normally distributed. The model of assumption was not confirmed by Shapiro-Wilk W Test (P value = 0.0338). *The data are not normally distributed.*

Day 3 (Screening Cut Point of Non-transformed Values):

Initial:

Distributions

Mean Response Without Drug



Normal(100.662,10.4423)

Quantiles

100.0%	maximum	134
99.5%		134
97.5%		134
90.0%		116.5
75.0%	quartile	104.75
50.0%	median	98.75
25.0%	quartile	92.875
10.0%		90
2.5%		87
0.5%		87
0.0%	minimum	87

Summary Statistics

Mean	100.66176
Std Dev	10.44228
Std Err Mean	1.7908363
Upper 95% Mean	104.30525
Lower 95% Mean	97.018281
N	34

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	100.66176	97.018281	104.30525
Dispersion	σ	10.44228	8.4224987	13.744941

-2log(Likelihood) = 255.006503760219

Goodness-of-Fit Test

Shapiro-Wilk W Test

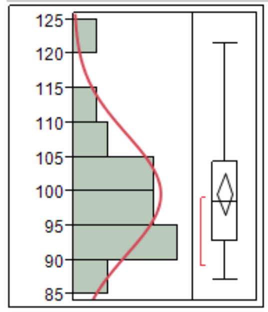
W	Prob<W
0.892861	0.0030*

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

Final:

Distributions

Mean Response Without Drug



Normal(99.6515,8.75579)

Quantiles

100.0%	maximum	121.5
99.5%		121.5
97.5%		121.5
90.0%		111.1
75.0%	quartile	104.25
50.0%	median	98.5
25.0%	quartile	92.75
10.0%		89.9
2.5%		87
0.5%		87
0.0%	minimum	87

Summary Statistics

Mean	99.651515
Std Dev	8.7557881
Std Err Mean	1.5241871
Upper 95% Mean	102.75618
Lower 95% Mean	96.546848
N	33

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	99.651515	96.546848	102.75618
Dispersion	σ	8.7557881	7.04131	11.58122

-2log(Likelihood) = 235.851132011183

Goodness-of-Fit Test

Shapiro-Wilk W Test

W	Prob<W
0.928461	0.0316*

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

The following individual MKSE lot was identified as an outlier:

Sample Number	Lot Number	Assay Date	Mean Response Without Drug	CV Response Without Drug (%)	Log Mean Response Without Drug
11	CYN141427	18Jul2014	134	0	2.1271

Data distribution was assessed using JMP Statistical Discovery with the assumption that the data is normally distributed. The model of assumption was not confirmed by Shapiro-Wilk W Test (P value = 0.0316). *The data are not normally distributed.*

The analysis of the non-transformed data show that 2 of 3 days are not normally distributed, therefore, the RU values were log transformed and evaluated for outliers and normality. Statistical outliers for screening using the $\log_{10}$ transformed RU values were evaluated on a per-day basis using the "box-plot" method in JMP software. Each exported Excel file is included for verification of outliers.

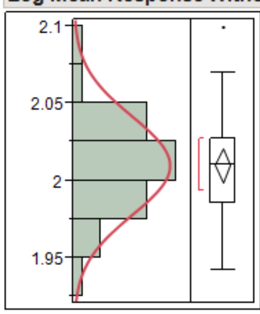
Next, the normality for screening analysis was assessed on a per-day basis using Shapiro-Wilk test in JMP software. The normality was indicated by the probability in the "Goodness-of-Fit". If the probability is greater and equal to 0.05, the data is normally distributed. If the probability is less than 0.05, the data is not normally distributed.

Day 1 (Screening Cut Point of Log Transformed Values):

Initial:

Distributions

Log Mean Response Without Drug



Normal(2.01001,0.03188)

Quantiles

100.0%	maximum	2.0986
99.5%		2.0986
97.5%		2.0986
90.0%		2.04764
75.0%	quartile	2.0273
50.0%	median	2.0107
25.0%	quartile	1.98565
10.0%		1.9689
2.5%		1.942
0.5%		1.942
0.0%	minimum	1.942

Summary Statistics

Mean	2.0100091
Std Dev	0.03188
Std Err Mean	0.0055496
Upper 95% Mean	2.0213133
Lower 95% Mean	1.9987049
N	33

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	2.0100091	1.9987049	2.0213133
Dispersion	σ	0.03188	0.0256376	0.0421675

-2log(Likelihood) = -134.771213120786

Goodness-of-Fit Test

Shapiro-Wilk W Test

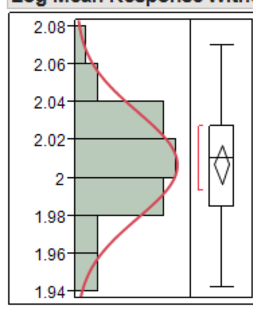
W	Prob<W
0.981896	0.8415

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

Final:

Distributions

Log Mean Response Without Drug



Normal(2.00724,0.02807)

Quantiles

100.0%	maximum	2.07
99.5%		2.07
97.5%		2.07
90.0%		2.04353
75.0%	quartile	2.0273
50.0%	median	2.0107
25.0%	quartile	1.98508
10.0%		1.9682
2.5%		1.942
0.5%		1.942
0.0%	minimum	1.942

Summary Statistics

Mean	2.0072406
Std Dev	0.028072
Std Err Mean	0.0049625
Upper 95% Mean	2.0173617
Lower 95% Mean	1.9971196
N	32

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	2.0072406	1.9971196	2.0173617
Dispersion	σ	0.028072	0.0225054	0.0373212

-2log(Likelihood) = -138.858790808605

Goodness-of-Fit Test

Shapiro-Wilk W Test

W	Prob<W
0.993017	0.9988

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

The following individual MKSE lot was identified as an outlier:

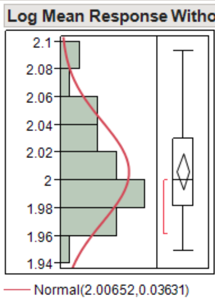
Sample Number	Lot Number	Assay Date	Mean Response Without Drug	CV Response Without Drug (%)	Log Mean Response Without Drug
11	CYN141427	16Jul2014	125.5	3.9	2.0986

Data distribution was assessed using JMP Statistical Discovery with the assumption that the data is normally distributed. The model of assumption was confirmed by Shapiro-Wilk W Test (P value = 0.9988). *The data are normally distributed.*

Day 2 (Screening Cut Point of Log Transformed Values):

Distributions

Log Mean Response Without Drug



Quantiles

100.0%	maximum	2.0934
99.5%		2.0934
97.5%		2.0934
90.0%		2.06234
75.0%	quartile	2.0304
50.0%	median	2
25.0%	quartile	1.98115
10.0%		1.96236
2.5%		1.9494
0.5%		1.9494
0.0%	minimum	1.9494

Summary Statistics

Mean	2.0065182
Std Dev	0.036305
Std Err Mean	0.0063199
Upper 95% Mean	2.0193914
Lower 95% Mean	1.993645
N	33

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	2.0065182	1.993645	2.0193914
Dispersion	σ	0.036305	0.0291961	0.0480204

-2log(Likelihood) = -126.192756157644

Goodness-of-Fit Test

Shapiro-Wilk W Test

W	Prob<W
0.947721	0.1141

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

Outliers were not present in the data from Day 2 of analysis.

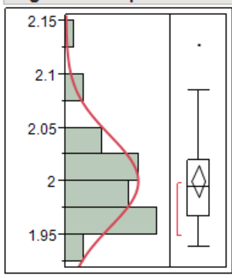
Data distribution was assessed using JMP Statistical Discovery with the assumption that the data is normally distributed. The model of assumption was confirmed by Shapiro-Wilk W Test (P value = 0.1141). *The data are normally distributed.*

Day 3 (Screening Cut Point of Log Transformed Values):

Initial:

Distributions

Log Mean Response Without Drug



Normal(2.00074,0.04282)

Quantiles

100.0%	maximum	2.1271
99.5%		2.1271
97.5%		2.1271
90.0%		2.06595
75.0%	quartile	2.02015
50.0%	median	1.9945
25.0%	quartile	1.9679
10.0%		1.9542
2.5%		1.9395
0.5%		1.9395
0.0%	minimum	1.9395

Summary Statistics

Mean	2.0007412
Std Dev	0.0428199
Std Err Mean	0.0073436
Upper 95% Mean	2.0156818
Lower 95% Mean	1.9858006
N	34

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	2.0007412	1.9858006	2.0156818
Dispersion	σ	0.0428199	0.0345375	0.0563629

-2log(Likelihood) = -118.763295688821

Goodness-of-Fit Test

Shapiro-Wilk W Test

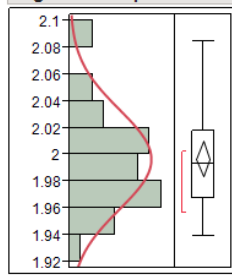
W	Prob<W
0.924244	0.0213*

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

Final:

Distributions

Log Mean Response Without Drug



Normal(1.99691,0.0371)

Quantiles

100.0%	maximum	2.0846
99.5%		2.0846
97.5%		2.0846
90.0%		2.04574
75.0%	quartile	2.01805
50.0%	median	1.9934
25.0%	quartile	1.9673
10.0%		1.95372
2.5%		1.9395
0.5%		1.9395
0.0%	minimum	1.9395

Summary Statistics

Mean	1.9969121
Std Dev	0.0371048
Std Err Mean	0.0064591
Upper 95% Mean	2.0100689
Lower 95% Mean	1.9837553
N	33

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	1.9969121	1.9837553	2.0100689
Dispersion	σ	0.0371048	0.0298393	0.0490783

-2log(Likelihood) = -124.754558837911

Goodness-of-Fit Test

Shapiro-Wilk W Test

W	Prob<W
0.946148	0.1027

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

The following individual MKSE lot was identified as an outlier:

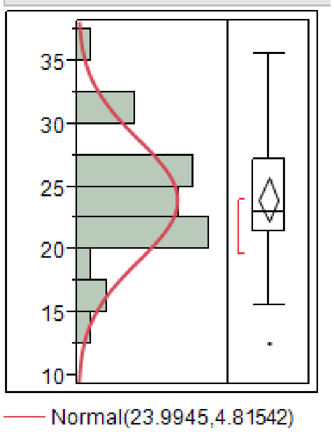
Sample Number	Lot Number	Assay Date	Mean Response Without Drug	CV Response Without Drug (%)	Log Mean Response Without Drug
11	CYN141427	18Jul2014	134	0	2.1271

Data distribution was assessed using JMP Statistical Discovery with the assumption that the data is normally distributed. The model of assumption was confirmed by Shapiro-Wilk W Test (P value = 0.1027). *The data are normally distributed.*

For the confirmatory cut point, statistical outliers for confirmatory data using percent inhibition were evaluated using the "box-plot" method in JMP software. The detailed formula is also listed in QPS SOP QPS-TLM-001. Individuals that have a response above the upper boundary or below the lower boundary were exported as an Excel file and excluded from further analysis. Each exported Excel file is included for verification of outliers.

Distributions

LT3114 Percent Inhibition



Quantiles

100.0%	maximum	35.5
99.5%		35.5
97.5%		35.5
90.0%		31.054
75.0%	quartile	27.115
50.0%	median	22.99
25.0%	quartile	21.39
10.0%		17.412
2.5%		12.5
0.5%		12.5
0.0%	minimum	12.5

Summary Statistics

Mean	23.994545
Std Dev	4.8154161
Std Err Mean	0.8382563
Upper 95% Mean	25.702018
Lower 95% Mean	22.287073
N	33

The following individual MKSE lot was identified as an outlier:

Sample Number	Lot Number	Assay Date	Mean Response Without Drug	Mean Response With LT3114 Drug	LT3114 Percent Inhibition (%)
31	CYN141445	16Jul2014	92	80.5	12.5

Appendix D Analytical Procedure for the Detection of Anti-LT3114 Antibodies in Monkey Serum by ECL Assay

Analytical Procedure for Detection of Anti-LT3114 Antibodies in Monkey Serum by ECL Assay

	Name	Signature	Date
PI Review and Approval	Breann Barker	Breann Barker	7/7/14
Scientific Review	Holly Shen	Holly Shen	7/7/14
Quality Assurance	Shengmei Ma	Shengmei Ma	7/10/14
Analytical Procedure Approval for Sample Analysis If Yes, then all validation data tables (i.e. except SSS & LTFS) must be compiled & signed off by PI			☐ YES ☒ NO

Copy made for QPS Project number	
Distribution Initials/Date*	
Copy Number	

*Analytical Procedure expires 7 days from the distribution date

QPS Study Number	
Client Study Number	
Run ID(s):	

1. OBJECTIVE

Sample Analysis (Check Type)	☐ Screening	☐ Confirmation	☐ Titration
Validation (List Test)			
Initial/ Date			

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2. MATERIALS AND EQUIPMENT

- 96 MSD Standard Coated Streptavidin plate: Cat# L11SA-2 or equivalent
- 25-100 mL Reagent reservoir, : Costar Cat.# 4870
- Plate sealer, Costar Cat.# 3095
- 96-well format micro titer/dilution tubes (VWR Cat.# 46300-536)
- Polyropylene dilution plate (Nunc Cat.# 267385)
- Pipette tips for single and multichannel pipettes
- Graduated cylinder: various volumes
- 8-Channel StatMatic II microplate dispenser

	Equipment Number		Equipment Number
Multi-tube Vortexer		20 µL Pipette	
Plate Shaker		100 µL Pipette	
Balance		200 µL Pipette	
Microplate reader: Meso Scale Discovery Sector Imager 6000		1000 µL Pipette	
		Multichannel Pipette(s)	
Initial/ Date			

3. SPECIFIC ASSAY MATERIALS, REAGENTS, ETC.

Not used materials must be marked as " NA"

Description	Lot Number	Expiration Date	Notebook
LT3114 (19.33 mg/mL) *LPath*	LP121416	4/30/16	Client 344 Compound Receipt NB
BSA, Fraction V (Fatty Acid Free) *(Calbiochem, Cat# 126575)*			NA
Biotinylated LT3114 (567 µg/mL)	BLB140609-02	6/9/16	Client 344 LT3114 Assay Notes
Sulfo-tag LT3114 (610 µg/mL)	BLB140609-03	6/9/16	Client 344 LT3114 Assay Notes
PBST (0.05% Tween 20 in 1X PBS)			
Streptavidin coated MSD Plate *(MSD Cat# L11SA-2 or equivalent)*			NA
4X MSD Read Buffer T *(MSD Cat# R92TC-1 or Equivalent)*			NA
Water, Type I, from Elgastat UHQ-PS, Elga Ltd. or equivalent			
Entered and Verified by (Initial/ Date):			

4. ASSAY SOLUTIONS

BUFFERS TO BE MADE DAILY:

Assay Buffer (1% BSA in PBST)
Circle appropriate scheme used

	1 Plate	2 Plates	3 Plates	4 Plates	See attached worksheet for alternative preparation
BSA	0.25 g	0.50 g	0.75 g	1.00 g	
PBST	25 mL	50 mL	75 mL	100 mL	
Total Volume	*25 mL*	*50 mL*	*75 mL*	*100 mL*	

Blocking Buffer (4% BSA in PBST)
Circle appropriate scheme used

	1 Plate	2 Plates	3 Plates	4 Plates	See attached worksheet for alternative preparation
BSA	0.64 g	1.28 g	1.92 g	2.56 g	
PBST	16 mL	32 mL	48 mL	28 mL	
Total Volume	*16 mL*	*32 mL*	*48 mL*	*64 mL*	

Dilution Buffer (3.3% Pooled MKSE in Assay Buffer): For titration analysis only
Circle appropriate scheme used

	1 Plate	2 Plates	3 Plates	4 Plates	See attached worksheet for alternative preparation
Pooled MKSE	117 µL	217 µL	300 µL	400 µL	
Assay Buffer	3.383 mL	6.283 mL	8.700 mL	11.600 mL	
Total Volume	*3.5 mL*	*6.5 mL*	*9.0 mL*	*12.0 mL*	

Initial/ Date	

5. POSITIVE AND NEGATIVE CONTROLS

Positive Controls (PC): Anti-LT3114 positive Rabbit Serum, Lot #: R-2265-B2-11/13, Exp Date: 5/21/16

Item	N=	Lot #	Exp Date	NB
HPC (1:500 in MKSE)				Client 344 LT3114 Assay Notes
MPC (1:2000 in MKSE)				
LPC (1:6000 in MKSE)				

Frozen	Storage temperature	
	Time removed from storage location	
	Time returned to storage location or check discarded if not returned	☐ Discarded

Negative Control (NC): Pooled Monkey Serum

Item	N=	Lot #	Exp Date	NB
Pooled Monkey Serum				

Initial/ Date	

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6. VALIDATION INFORMATION

<table>
<tr><th>Validation Test</th><th>N=</th><th colspan="4">Instruction</th></tr>
<tr><td>Determination of Screening Cut Point</td><td></td><td colspan="4">☐ Refer to attached sheet for individual lot numbers.</td></tr>
<tr><td>Determination of Confirmatory Assay Cut Point</td><td></td><td colspan="4">☐ Refer to attached sheet for individual lot numbers.</td></tr>
<tr><td>Drug Tolerance Test</td><td></td><td colspan="4">☐ See attached worksheet for sample preparation</td></tr>
<tr><td rowspan="6">Room Temperature Stability</td><td rowspan="6"></td><td>Time:</td><td colspan="3"></td></tr>
<tr><td>PCs in Monkey Serum</td><td>Lot #</td><td>Exp. Date</td><td>NB</td></tr>
<tr><td>HPC (1:500)</td><td></td><td></td><td rowspan="3">Client 344 LT3114 Assay Notes</td></tr>
<tr><td>MPC (1:2000)</td><td></td><td></td></tr>
<tr><td>LPC (1:6000)</td><td></td><td></td></tr>
<tr><td colspan="4">☐ Refer to QPS Stability Forms for more information</td></tr>
<tr><td rowspan="6">Freeze Thaw Cycle Stability (-70°C)</td><td rowspan="6"></td><td>Cycle:</td><td colspan="3"></td></tr>
<tr><td>PCs in Monkey Serum</td><td>Lot #</td><td>Exp. Date</td><td>NB</td></tr>
<tr><td>HPC (1:500)</td><td></td><td></td><td rowspan="3">Client 344 LT3114 Assay Notes</td></tr>
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<tr><td>LPC (1:6000)</td><td></td><td></td></tr>
<tr><td colspan="4">☐ Refer to QPS Stability Forms for more information</td></tr>
<tr><td>Matrix Selectivity (Recovery) Test</td><td></td><td colspan="4">☐ See attached worksheet(s) for sample preparation and individual human serum lots</td></tr>
<tr><td colspan="3">Initial/Date</td><td colspan="3"></td></tr>
</table>

7. SAMPLE INFORMATION

Sample Storage *(Refer to form 177 in study notebook for further details)*	
Length of established Room Temperature Stability	To be determined
Number of Samples	
Initial/Date	

Sample Sequence Verification *(Must be performed by another staff member)*	**Initial/Date**	

Re-analysis Sample Summary

Sample IDs:		☐ See attached worksheet for list of samples
Reason:		
Management authorization for client requested re-assay		
Initial/Date		

8. ASSAY PROCEDURE

Note: Analyst must have been trained for ECL assays and hand/machine washing technique. All plate incubations are performed on plate shaker (settings 3-4) at room temperature unless otherwise specified.

Step	Action	Initial/ Date
1	**Block Plate** Block plate with 150 µL of Blocking Buffer and incubate for at least an hour.	

	1A	2B	3C	4D
Start:				
End:				
MSD Barcode:				

Step	Action	Initial/ Date
2	**Sample Preparation (Screening)** **a)** Arrange dilution tubes according to attached plate map. Use one tube for duplicate measurement of one sample. **b)** Prepare a 1:30 dilution of each study sample/NC/PC in Assay Buffer: *10 µL study sample/NC/PC + 290 µL* ***Assay Buffer*** **c)** Proceed to Step 5	

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Step	Action	Initial/ Date
3	**Sample Preparation (Confirmation)** **a)** Arrange dilution tubes according to attached plate map. Use one tube for duplicate measurement of one sample. **b)** Prepare LT3114 Stock B (2 mg/mL): *Circle appropriate scheme:*	

	1 Plate	2 Plates	3 Plates	4 Plates	See attached worksheet for alternative preparation
LT3114 Stock A **(19.33 mg/mL)**	12.9 µL	23.3 µL	33.6 µL	44.0 µL	
Pooled HUSE	112.1 µL	201.7 µL	291.4 µL	381.0 µL	
Total Volume	*125 µL*	*225 µL*	*325 µL*	*425µL*	

c) Spike each study sample/ HPC / LPC with 100 µg/mL LT3114:

5 µL LT3114 Stock B (2000 µg/mL) + 95 µL PC or sample

d) Incubate spiked samples for at least 30 minutes at room temperature.

	1A	2B	3C	4D
Start:				
End:				

e) Dilute spiked and un-spiked samples and PCs 1:30 in Assay Buffer:

10 µL sample + 290 µL ***Assay Buffer***

f) Proceed to Step 5

Step	Action	Initial/ Date
4	**Prepare Master Mix:** **a)** Prepare 20 µg/mL Biotinylated LT3114 Stock B: Add 12 µL Biotinylated LT3114 Stock A (567 µg/mL) + 328.2 µL Assay Buffer **b)** Prepare 20 µg/mL Sulfo-tag LT3114 Stock B: Add 12 µL Sulfo-tag LT3114 Stock A (610 µg/mL) + 354 µL Assay Buffer **c)** Prepare 250 ng/mL Biotinylated LT3114 and Sulfo-tag LT3114 Master Mix: *Circle appropriate scheme:*	

	1 Plate	2 Plates	3 Plates	4 Plates	See attached worksheet for alternative preparation
Biotinylated LT3314 **(20 µg/mL)**	75 µL	150 µL	200 µL	275 µL	
Sulfo-tag LT3114 **(20 µg/mL)**	75 µL	150 µL	200 µL	275 µL	
Assay Buffer	5.85 mL	11.7 mL	15.6	21.45 mL	
Total Volume	*6 mL*	*12 mL*	*16 mL*	*22 mL*	

Step	Action	Initial/ Date
5	**Sample Loading to Polypropylene Plate:** **a)** Add 50 µL/well of each diluted sample/NC/PC according to attached plate map. **b)** Add 100 µL/well of Master Mix. *(Note: Final volume of each well in polypropylene plate is sufficient for duplicate measurement when transferred to MSD plate in Step 7)* **c)** Cover plate with a plate sealer and incubate **1 hour** ± **10 min** in the dark at room temperature on a plate shaker (setting 3-4).	
6	**Sample Addition:** **a)** Wash blocked MSD plate 3X with PBST. **b)** Transfer 50 µL of the incubated sample mixture from the polypropylene plate into the MSD plate according to plate map. **c)** Cover plate with a plate sealer and incubate for **1-1.5 hours** in the dark at room temperature on a plate shaker (setting 3-4).	
7	**Dilute 4X Read Buffer to 2X using type I water:** *Circle appropriate scheme used*	
8	**Read Plate:** **a)** Wash plate **3X** with PBST. **b)** Add **150 µL/well** 1X Read Buffer from step 8.8 and mix thoroughly by tapping the plate. **c)** Read plate on **MSD Sector Imager 6000**. Name the file after the run ID: "RunID.txt". **Save file.** ☐ Printscreen of the MSD raw data attached to procedure ☐ Plate layout template attached to procedure	

Step 5 incubation times:

	1A	2B	3C	4D
Start:				
End:				

Step 6 incubation times:

	1A	2B	3C	4D
Start:				
End:				

Step 7 dilution schemes:

	1 Plate	**2 Plates**	**3 Plates**	**4 Plates**	See attached worksheet for alternative preparation
4x Read Buffer	8 mL	16 mL	24 mL	32 mL	
Type I H_2O	8 mL	16 mL	24 mL	32 mL	
Total Volume	*16 mL*	*32 mL*	*48 mL*	*64 mL*	

Step	Action	Initial/ Date
9	**Data Processing:** Reduction: ☐None ☐ Data transferred to Watson ☐ other _________	

9. WORK FLOW

Analyst(s) performing ELISA:	
Peer Review	

10. PRINCIPAL INVESTIGATOR/STUDY DIRECTOR APPROVAL

Determine pass/fail of each analytical run according to the acceptance criteria specified in the QPS Protocol 344-1404.

Run Number	Run Pass (P)/Fail(F) (if F, complete section below)

Do test(s) during the validation meet the acceptance criteria as specified in the SOPs (e.g. Stability Test(s), Recovery, Specificity, Other)?	☐ Yes ☐ No	If no, complete following page
Should Study Samples be re-assayed? Decision is determined from the data interpreted today.	☐ Yes ☐ No	If yes, List sample identification and reason or reassay:
Principal Investigator (Initial/Date)		

Review the following to:

☐ Determine root cause of failed run or test (if applicable)

☐ Investigate experimental observations:____

☒ = Reviewed & evaluated ☐ = Not applicable (NA)

☐ Analytical Procedure Documentation:	☐ Washing method:
☐ Standard /QC Preparation Documentation:	☐ Conjugate preparation and incubation:
☐ Buffer Solution Preparation:	☐ Plate read setup:
☐ Plate preparation (lot#, Ab, #days, etc.):	☐ Analyst laboratory Error:
☐ Matrices:	☐ Equipment maintenance log:

Conclusion:

Principal Investigator (Initial/Date)	